Trainer : MOHAN					Batch : AIML_003
				Time : 11 to 1 PM - March 28th	Days : Saturday & Sunday(weekend)
Week	**Class**			**Major Topics**	**Subtopic Covered**
Week 1	Day 1			AI & ML Overview	-Definitions, Use Cases, Lifecycle
	Day 2	04/04/2026	Saturday	Python Basics, Functions	-Python Setup -Variables, Numbers, Strings -Lists, Dictionaries, Sets, Tuples
Week 2	Day 3	05/04/2026	Sunday	Python	- Condition, loops - Functions, Lambda Functions
	Day 4	11/04/2026	Saturday	Python	Read, Write files Exception handling Classes, Objects
Week 3	Day 5	12/04/2026	Sunday	Numpy , Pandas	NumPy Arrays and Functions Pandas DataFrames Creating, Accessing, Modifying, and Combining DataFrames Pandas Functions Various Data Manipulation Functions in Pandas Saving and Loading Datasets Using Pandas
	Day 8	25/04/2026	Saturday	Maths (Linear Algebra)	Linear algebra: Vectors, matrices, inner products, matrix-vector multiplication, eigen values/vectors, singular value decomposition
Week 5	Day 9	26/04/2026	Sunday	Maths (Probability, statistics)	Probability Theory: Discrete and continuous random variables, probability distributions, Bayes' rule. Statistics: Descriptive and inferential statistics, hypothesis testing, probability distributions
	Day 10	02/05/2026	Saturday	EDA	Data Overview General Statistics of the Dataset (e.g., describe(), info()) Univariate Analysis (Histogram, Boxplots, and Bar Graphs) Bivariate/Multivariate Analysis (Line Plot, Scatterplot, Lmplot, Jointplot, Violin Plot, Strip Plot, Swarmplot, Catplot, Pairplot, Heatmap)
Week 6	Day 11	03/05/2026	Sunday	ML Intro	Missing Value Treatment Outlier Detection and Treatment Scikit-learn Basic ML workflow
Week 7	Day 12	09/05/2026	Saturday	ML algorithm	Feature Engineering (Preprocessing) Feature Selection,
	Day 13	10/05/2026	Sunday	ML algorithm	Feature scalling , One hot encoding, label encoding Train test split Cross validation
Week 8	Day 14	16/05/2026	Saturday	ML algorithm	Regression Linear
	Day 15	17/05/2026	Sunday	ML algorithm	Logistic Model evaluation Metrics
Week 9	Day 16	23/05/2026	Saturday	ML Adv Algorithm	Decision Trees
	Day 17	24/05/2026	Sunday	ML Adv Algorithm	Random Forest
Week 10	Day 18	30/05/2026	Saturday	ML Adv Algorithm	KNN Naive Bayes
	Day 19	31/05/2026	Sunday	ML Adv Algorithm	SVM
Week 11	Day 20	06/06/2026	Saturday	ML Adv Algorithm	Clustering PCA
	Day 21	07/06/2026	Sunday	ML	Hyperparameters Cross-validation
Week 12	Day 22	13/06/2026	Saturday	ML Adv Algorithm	XGBoost

	Day 23	14/06/2026	Sunday	Deep Learning	What is a neural network? Forward propagation Back propagation
Week 13	Day 24	20/06/2026	Saturday	Deep Learning	Perceptrons MLP Building multilayer perceptron
	Day 25	21/06/2026	Sunday	Deep Learning	Activation Functions Sigmoid , ReLU , Softmax weights & biases
Week 14	Day 26	27/06/2026	Saturday	Deep Learning	Keras/PyTorch Tensors and Operations
	Day 27	28/06/2026	Sunday	Deep Learning	Building Neural Networks ANN
Week 15	Day 28	04/07/2026	Saturday	Deep Learning	CNN Use cases
	Day 29	05/07/2026	Sunday	Deep Learning	Model Training Training Neural Networks Pre-trained Models
Week 16	Day 30	11/07/2026	Saturday	Deep Learning	Train an MLP/CNN on an open dataset
	Day 31	12/07/2026	Sunday	Transformers & Attention	Self-attention Introduction to Transformers Components of a Transformer
Week 17	Day 32	18/07/2026	Saturday	Transformers & Attention	Encoder architecture decoder architecture Different Transformer Architectures
	Day 33	19/07/2026	Sunday	Transformers & Attention	Applications of Transformers Hugging Face library basics
Week 18	Day 34	25/07/2026	Saturday	LLM	Introduction to LLMs Working of LLMs Applications of LLMs
	Day 35	26/07/2026	Sunday	Prompt Eng	Introduction to Prompt Engineering Strategies for Devising Prompts
Week 19	Day 36	01/08/2026	Saturday	Embeddings & Tokenization	Introduction to Embeddings and Tokenization Byte-Pair Encoding (BPE) Tokenization
	Day 37	02/08/2026	Sunday	Embeddings & RAG	Computation and Application of Sentence Embeddings Retrieval-Augmented Generation (RAG)
Week 20	Day 38	08/08/2026	Saturday	Vector Stores & RAG Systems	Embeddings, Vector stores Build a RAG-based chatbot using LLMs
	Day 39	09/08/2026	Sunday	LLM Tooling & Platforms	OpenAI AWS Bedrock
Week 21	Day 40	15/08/2026	Saturday	Fine-tuning Fundamentals	Fine-tune GPT models Fine-tuning
	Day 41	16/08/2026	Sunday	Parameter-Efficient Fine-Tuning	Parameter-Efficient Fine-Tuning (PEFT) PEFT Techniques (Prefix Tuning, Prompt Tuning, QLoRA)
Week 22	Day 42	22/08/2026	Saturday	QLoRA	QLoRA Application and Implementation
	Day 43	23/08/2026	Sunday	Agent Systems	Agent frameworks
Week 23	Day 44	29/08/2026	Saturday	Diffusion Models Basics	Diffusion image/audio synthesis
	Day 45	30/08/2026	Sunday	Diffusion Applications	Audio Noise Distillation DSP pipelines Build a diffusion-based image generator
Week 24	Day 46	05/09/2026	Saturday	MLOps Fundamentals	What is ML Ops? Experiment Tracking with MLFlow
	Day 47	06/09/2026	Sunday	DevOps for ML	DevOps Fundamentals: CI/CD pipelines containerization (Docker, Kubernetes)

Week 25	Day 48	12/09/2026	Saturday	Cloud & Orchestration	Familiarity with at least one cloud platform (AWS, Azure etc.) Kubeflow
	Day 49	13/09/2026	Sunday	Pipelines & Lifecycle	Pipelines Basic Workflows Experiment Tracking
Week 26	Day 50	19/09/2026	Saturday	Model Lifecycle & Packaging	Packaging Code and Running Projects Model Lifecycle Management
	Day 51	20/09/2026	Sunday	Deployment & Monitoring	FastAPI Deployment Model Monitoring
Week 27	Day 52	26/09/2026	Saturday	Frontend Tools	Gradio Streamlit Basics
	Day 53	**27/09/2026**	Sunday	Full AI Application	Backend + Frontend integration Deploy a full AI app with frontend + backend integration
			Final discussion		